iTech's Magical Differentiated OCP Tracker

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I. LIS 5367 Final Project Write Up

My project is a web-based application that tracks student skills in a post-secondary vocational education program. It maps computer-based assignments with the curriculum frameworks provided by the Florida Department of Education. Students can see their progress and mastery. The app also provides a mechanism for recording student skills observed during work-based activities. This allows for differentiated instruction accommodating students who are English language learners or those with disabilities.

The original website and database were hosted on a local Ubuntu VM using LAMPP. To increase availability of the project, I migrated the web server and PHP to an AWS Amazon Linux (a proprietary "Amazon" OS) EC2 instance and transferred the MySQL database to the Amazon RDS service. The goal was to decouple as much of the application as possible. While the core application is hosted in EC2, the database is hosted in Amazon RDS. Fault tolerance can also be set up in FSU's Torch.

Code is written in HTML, PHP, MySQL and minimal CSS. Versioning was done with Git and synced with GitHub. For fun, I used an API to call a random programming joke at the bottom of the home page. All of these are new to me since I come from the infrastructure side of the house. My goal was to try several different technologies and I enjoyed the process thoroughly. I found that the more I got into the project, the more functionality I wanted to add.

I am looking forward to continuing developing this project and learning more technologies. Some of these future goals include:

- Hardening the application:
  - Validation of input from forms
  - Sanitization of SQL
  - Restricting access by user
  - Pass-through authentication from Canvas using OAuth2 (API)
- Additional reporting.
- Incorporating Python and JavaScript for advanced functionality.
- Automate syncing with Canvas.
- Improve design of the page.

II. The Code

- GitHub: https://github.com/AJMFSU/LIS5367
  (.sql file can be found in the data directory)

III. The Demo

To access the project, open a browser and navigate to https://apps.themcmullin7.com

Task 1: You are a student who wants to see your progress, but you have forgotten your student number.

Steps:

1. From the home page, look up your student number by clicking on “Student Lookup”
2. Enter “caleb” in the first name box and click submit
   (you can also lookup “mcmullin”, “juan”, “344124” or click on the link to show all students)
3. Make note of the student number for Caleb.
4. Return to the home page by clicking the iTech logo in the upper right.
5. Click on “check my progress.”
6. Enter the student number from step 3 and view the results.
   (You can also look up 148834, or 344124)
7. Return to the home page by clicking the iTech logo in the upper right.

NOTE: This application and set of tasks has been written for LIS5367 and is optimized to demonstrate functionality. A real-world implementation would not allow unauthorized users to lookup student information.

Task 2: You are a teacher and you want to enter live-work for a student.

Steps:

1. Click on “Enter Observed Skills”
2. Complete form and hit submit.
3. Confirm that you have awarded a standard by clicking “View student progress” on the confirmation page.

Feel free to explore other links in the program. Thank you.